

REPRO_RL_urafi3

urafi3@gatech.edu

1. Overleaf Read-Only Link

<https://www.overleaf.com/read/bmdkxrkjfgbq#412ff5>

2. GitHub Commit Hash

718e3a833bcb34299f97a6031684a140edb2c885

3. Environment Setup

Tested on Windows 11, Python 3.10+. Run the following to set up:

```
git clone https://github.gatech.edu/YOUR-GTUSERNAME/YOUR-REPO.git
cd YOUR-REPO
pip install -r requirements.txt
```

4. Run Instructions

All experiments are launched from a single entry point:

```
cd rl/src
python run_all.py
```

This will:

- Build the Blackjack and CartPole transition models
- Run VI and PI on both environments
- Run SARSA and Q-Learning on both environments across 5 seeds {42, 7, 13, 99, 2024}
- Run the CartPole discretization ablation over 5 grids
- Generate all 14 figures and save them to output/figures/

5. Output

All figures are saved to:
output/figures/

6. Random Seeds

All experiments use seeds {42, 7, 13, 99, 2024} set via `numpy.random.default_rng(seed)` at the start of each training run. Environment resets also use the seed stream. Seeds are defined in `rl/src/config.py` under SEEDS.

7. Key Configuration

All hyperparameters are centralized in `rl/src/config.py`. The champion settings used in the report are:

Parameter	Value
γ	0.99
α_0	0.5
$\alpha_{\min}$	0.1
ϵ_0	1.0
$\epsilon_{\min}$	0.01
T_{ϵ} (steps)	10,000
Episodes	5,000
CartPole grid	(3, 3, 8, 12)

8. File Structure

```
rl/  
  src/  
    config.py          # all hyperparameters and seeds  
    run_all.py         # main entry point  
    vi_pi.py          # Value Iteration and Policy Iteration  
    sarsa.py           # SARSA  
    qlearning.py       # Q-Learning  
    discretize.py      # CartPole binning and transition model  
    evaluate.py        # post-training evaluation  
    hyperparameter_search.py # staged search  
    plots.py          # all figure generation  
  output/  
    figures/           # generated PNGs
```